Summary of Fourier Series for $f(x)$ with Period $P = 2L$		
$f(x)$	Fourier Coefficients	Fourier Series
even	$d_n = \dfrac{1}{L}\displaystyle\int_0^L f(x)\cos\left(\dfrac{\pi n x}{L}\right)dx$	$f(x) = \displaystyle\sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L}$ $= d_0 + \displaystyle\sum_{n=1}^{\infty} 2\, d_n \cos\left(\tfrac{\pi n x}{L}\right)$
odd	$d_n = \dfrac{-i}{L}\displaystyle\int_0^L f(x)\sin\left(\dfrac{\pi n x}{L}\right)dx$	$f(x) = \displaystyle\sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L}$ $= \displaystyle\sum_{n=1}^{\infty} 2\, i\, d_n \sin\left(\tfrac{\pi n x}{L}\right)$
neither	$d_n = \dfrac{1}{2L}\displaystyle\int_{-L}^{L} f(x)\, e^{-i\pi n x/L}\, dx$	$f(x) = \displaystyle\sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L}$ $= d_0 + \displaystyle\sum_{n=1}^{\infty} [d_n + d_{-n}]\cos\left(\tfrac{\pi n x}{L}\right) + \displaystyle\sum_{n=1}^{\infty} [d_n - d_{-n}] i \sin\left(\tfrac{\pi n x}{L}\right)$